

Ditsa Chakraborty

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PROFILE SUMMARY

Electronics & Communication Engineering student with hands-on experience across AI/ML, data science, embedded systems, and software development. Took ownership of building and deploying production fintech applications during an industry internship, handling everything from data analysis to CI/CD deployment. Also worked on ESP32 based embedded communication systems and validated ML models (Neural Networks, Random Forest, CNN based computer vision) at 90%+ accuracy.

EDUCATION

B.Tech., Electronics & Communication Engineering

SRM Institute of Science and Technology

CGPA: 8.34 / 10 (up to 6th Semester)

2023 – 2027

Chennai, TN

Std XII — CBSE

Indus Valley World School

75.2%

2022

Kolkata, WB

Std X — CBSE

Indus Valley World School

86%

2020

Kolkata, WB

INDUSTRY EXPERIENCE

InvestTech Analyst Intern

ZeTheta Algorithms Pvt. Ltd.

02/2026 – 05/2026

Remote

- Owned end-to-end delivery of four production fintech applications (Market Depth Explorer, Convertible Bond Pricing Framework, Market Monitoring Software, ML-Based Sentiment Analysis Trading Platform) from requirements through deployment, coordinating across design, engineering, and QA.
- Drove process improvements in the model validation pipeline, reducing turnaround time on statistical model checks and improving reliability of trading signal outputs.
- Built and maintained CI/CD deployment workflows (Render, Vercel) and Git-based version control processes adopted across the intern team, improving release consistency and reducing deployment errors.
- Performed quantitative analysis and statistical modeling using Python and Pandas; built an NLP pipeline using FinBERT to analyze sentiment, generate trading signals, and run simulations.
- Delivered full ownership transfer of all four repositories to the company, including a recorded walkthrough and demo of each project (linked via unlisted YouTube video in the README) to support handoff.

VOLUNTEER EXPERIENCE

Volunteer, My Children Foundation (NGO)

06/2025 – 07/2025

Kolkata, WB

- Contributed to digital outreach initiatives supporting underprivileged children, including editing and transcribing videos for social media and fundraising campaigns.

ACADEMIC PROJECTS

Transmission and Reception of Signal Using ESP-32 with Machine Learning

SRM Institute (Team)

05/2026

Chennai, TN

- Built and deployed a multi-nodal communication system using ESP32 devices for signal transmission and reception experiments.
- Conducted performance experiments to gather and analyze RSSI data under various conditions and beams.
- Used Neural Network and Random Forest classification algorithms to predict communication behavior with 99.7% and 90.65% accuracy, respectively.
- Built a real-time Python dashboard for tracking communication data.
- Presented work at RASET 2026 – a National Conference organized by IEEE OES.

Fruit Quality Assessment using MobileNetV2

SRM Institute

04/2026

Chennai, TN

- Designed and implemented a Convolutional Neural Network (CNN) machine vision pipeline using MobileNetV2 with transfer learning for classifying fresh vs. spoiled fruit, achieving 96.1% accuracy.
- Conducted data pre-processing, model training, testing, and validation; completed technical documentation and material for an IEEE publication.

TECHNICAL SKILLS

Data Analytics & Machine Learning: Pandas, Scikit-Learn, Data Analysis, Data Visualization, Plotly, Neural Networks, Random Forest, MobileNetV2, FinBERT, NLP

Embedded Systems & ECE: ESP32, Arduino, FPGA/Verilog, IoT, LTspice, Scilab

Programming Languages: Python, SQL, Java, TypeScript, JavaScript

Databases & Cloud: PostgreSQL, Supabase, Vercel, Render

Software Development: React, Node.js, Express.js, REST APIs, Git, CI/CD
Operations & Process: Cross-functional Coordination, Process Optimization, Project Execution
Professional Skills: Research, Technical Documentation, Problem Solving, Presentation Skills, Team Collaboration, Design Thinking

CERTIFICATIONS & COURSES

Certificate of Presentation — RASET 2026 (IEEE OES) <i>Bannari Amman Institute of Technology</i>	05/2026
Python for Data Science, NPTEL <i>IIT Madras</i>	07/2024 – 08/2024
AI Foundations for Everyone, Coursera <i>IBM</i>	04/2025
Design Thinking – A Primer, NPTEL <i>IIT Madras</i>	01/2025 – 02/2025
Certificates of Project Work <i>Phantom Liquidity Hunter, Convertible Bond Laboratory, Market Depth Explorer, Sentiment Analysis Trading — ZeTheta Algorithms</i>	03/2026 – 05/2026

ACHIEVEMENTS

First Prize — Cooking Without Fire, 15th Annual Cultural Competition	2024
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